

Project 2.5: English-Portuguese Machine Translation

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Abstract

Neural Machine Translation (NMT) has become the dominant approach for machine translation due to its ability to learn end-to-end mappings between languages. In this project, we investigate English-Portuguese translation using the Europarl corpus. Rather than proposing a novel model, the objective is to experimentally evaluate different preprocessing strategies and Seq2Seq variants. Specifically, we compare hyperparameter settings, embedding initialization methods, translation directions, character-level modeling, attention mechanisms, and pivot translation. Translation quality is evaluated using BLEU and chrF.

1 Introduction

Neural Machine Translation (NMT) has become the dominant approach for automatic translation by learning end-to-end mappings between source and target languages using neural networks [2]. Compared with traditional statistical methods, NMT is capable of capturing long-range contextual information and generating more fluent translations [1]. In this project, we investigate English-Portuguese machine translation using the Europarl parallel corpus. Rather than focusing on theoretical aspects, the project emphasizes experimental evaluation. We first explore and preprocess the dataset, followed by the implementation of a recurrent sequence-to-sequence model. Several experiments are conducted to study the effects of hyperparameter choices, embedding initialization, translation direction, and character-based modeling. The baseline model is further extended with an attention mechanism, and an additional pivot translation system is developed for Portuguese-to-Swedish translation using English as the intermediate language. Model performance is evaluated using BLEU and chrF, and the experimental results are analyzed to compare different approaches.

2 Experimental Setup

In this section, we will explain the dataset source, our preprocessing methods, the design and reasoning for model architecture and our evaluation metrics.

2.1 Dataset

The experiments were conducted on the Europarl v7 English-Portuguese parallel corpus, which contains aligned sentence pairs from the proceedings of the European Parliament. Following the project requirements, 10 percent of the corpus was randomly sampled for all experiments. The sampled data were split into training, validation, and test sets, with 20 percent reserved for testing. Basic corpus statistics, including sentence length distributions and vocabulary sizes, are presented in Section 3.1.

2.2 Preprocessing

Several preprocessing steps were applied to improve data quality while preserving sentence alignment. Empty sentence pairs were

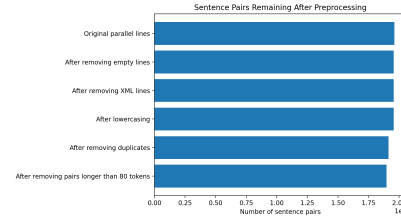


Figure 1: Number of sentence pairs remaining after each preprocessing step.

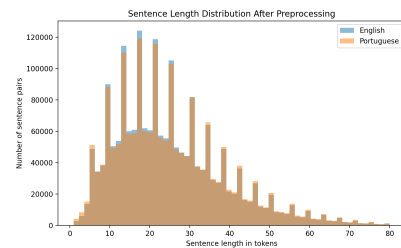


Figure 2: Sentence length distributions after preprocessing.

removed, XML metadata lines were discarded, and all text was converted to lowercase. Duplicate sentence pairs were then removed to reduce redundancy in the training data. Finally, sentence pairs containing more than 80 tokens in either language were excluded to limit sequence length and reduce computational cost.

After preprocessing, the corpus was reduced from 1,960,407 to 1,897,371 aligned sentence pairs, retaining over 96% of the original data. Figure 1 illustrates the number of remaining sentence pairs after each preprocessing step.

2.3 Model Architecture

A sequence-to-sequence (Seq2Seq) neural machine translation model was implemented using an LSTM-based encoder and decoder. Both components use embedding layers to represent source and target words, with support for either randomly initialized or pre-trained GloVe embeddings. During training, teacher forcing was applied to improve convergence, and dropout was used for regularization. The model was optimized using the Adam optimizer with the cross-entropy loss function while ignoring padding tokens. To further improve translation quality, the baseline Seq2Seq model was extended with a Bahdanau attention mechanism, allowing the decoder to attend to relevant encoder hidden states at each decoding step.

2.4 Evaluation Metrics

Model performance was evaluated using the **BLEU** and **chrF** metrics. BLEU measures translation quality based on the precision of matching word n -grams between the generated translation and

the reference sentence, while applying a brevity penalty to discourage overly short translations. In contrast, chrF computes the F-score over character n -grams, making it more robust to morphological variations and suitable for evaluating translations between morphologically rich languages such as English and Portuguese.

During training, the model was optimized using the cross-entropy loss, and the checkpoint with the lowest validation loss was selected for final evaluation. BLEU and chrF scores were then computed on the held-out test set to compare different model configurations.

3 Experimental Results

In this section, we will present the results we achieved from experiments which we explained in the section 2.

3.1 Data Exploration

The Europarl English–Portuguese corpus contains approximately 1.96 million aligned sentence pairs. After removing empty sentence pairs, over 1.95 million aligned pairs remained for subsequent pre-processing and model training.

Figure 3 shows the sentence length distributions of both languages. Most sentences contain fewer than 60 tokens, with average lengths of 25.17 tokens for English and 25.52 tokens for Portuguese. Portuguese sentences are slightly longer on average, although the overall distributions are highly similar.

Figure 4 illustrates the relationship between English and Portuguese sentence lengths. A strong positive correlation is observed, indicating that longer English sentences generally correspond to longer Portuguese translations. Only a few outliers with exceptionally long sentences are present.

Vocabulary statistics also reveal that Portuguese has a substantially larger vocabulary than English (135,751 vs. 83,837 unique tokens), reflecting its richer morphological structure. These observations suggest that the two languages are well aligned while Portuguese presents slightly greater lexical diversity, which may increase the difficulty of translation.

3.2 Hyperparameter Tuning

Manual hyperparameter tuning was performed to determine a suitable baseline configuration for the Seq2Seq model. Three model configurations were evaluated by varying the embedding dimension, hidden dimension, and dropout rate, while keeping the remaining training settings unchanged.

Table 1 summarizes the results. The model with a 128-dimensional embedding, a 256-dimensional hidden state, and a dropout rate of 0.3 achieved the best overall performance, obtaining the highest BLEU (0.229) and chrF (10.22) scores. Increasing the model size to 256-dimensional embeddings and 512 hidden units did not improve translation quality. Furthermore, increasing the dropout rate from 0.3 to 0.5 led to a substantial decrease in both BLEU and chrF, indicating that excessive regularization hindered learning.

Based on these results, the configuration with 128-dimensional embeddings, 256 hidden units, and 0.3 dropout was selected as the baseline model for all subsequent experiments.

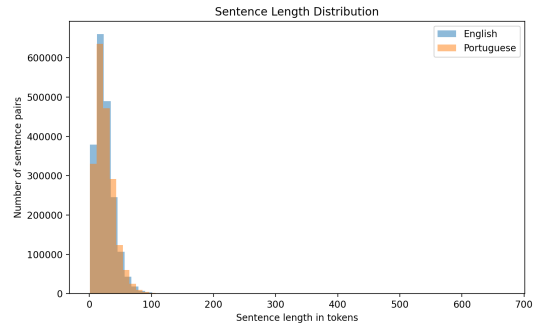


Figure 3: Sentence length distributions for English and Portuguese.

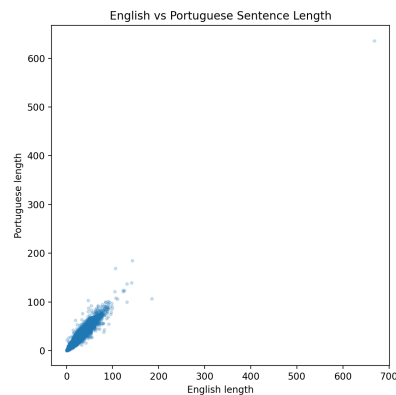


Figure 4: Relationship between English and Portuguese sentence lengths.

Table 1: Manual hyperparameter tuning results.

Configuration	Emb.	Hidden	Dropout	BLEU	chrF
Small	128	256	0.3	0.229	10.22
Base	256	512	0.3	0.226	10.18
Base + Dropout	256	512	0.5	0.082	5.15

3.3 Embedding Comparison

To evaluate the effect of embedding initialization, three embedding strategies were compared: randomly initialized embeddings, pre-trained GloVe-100 embeddings, and pre-trained GloVe-300 embeddings. All other model architecture and training settings were kept unchanged to ensure a fair comparison.

Figure 5 and Figure 6 present the BLEU and chrF scores for the three embedding models. The GloVe-300 embeddings achieved the best performance on both metrics, obtaining the highest BLEU and chrF scores. In contrast, GloVe-100 performed slightly worse than randomly initialized embeddings. This suggests that higher-dimensional pre-trained embeddings provide richer semantic representations, which help the encoder learn more informative sentence representations. Overall, the results indicate that embedding quality

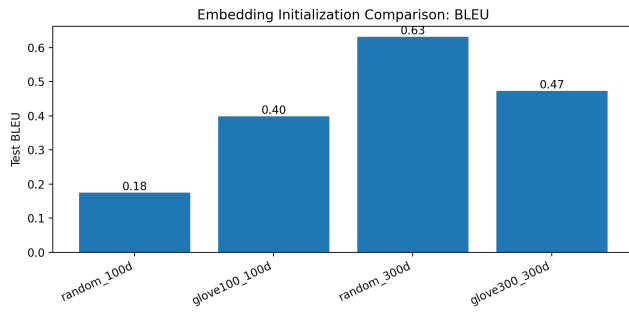


Figure 5: BLEU scores obtained with different embedding models.

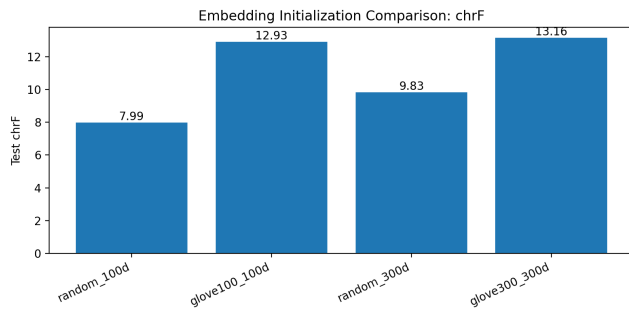


Figure 6: chrF scores obtained with different embedding models.

Table 2: Comparison of translation directions.

Direction	BLEU	chrF
EN→PT	0.640	12.14
PT→EN	1.066	14.33

has a noticeable impact on translation performance, with GloVe-300 being the most effective choice for the subsequent experiments.

3.4 EN→PT vs PT→EN

To investigate the effect of translation direction, the baseline Seq2Seq model was trained for both English-to-Portuguese (EN→PT) and Portuguese-to-English (PT→EN) translation using the same network architecture and training settings.

Table 2 summarizes the experimental results. The PT→EN model achieved higher BLEU (1.066) and chrF (14.33) scores than the EN→PT model, which obtained a BLEU score of 0.640 and a chrF score of 12.14. These results indicate that, under the current experimental setting, translating from Portuguese to English was easier for the Seq2Seq model. One possible reason is that English has a simpler morphological structure and a smaller target vocabulary, making it easier for the decoder to generate accurate translations. Overall, the translation direction has a noticeable impact on model performance.

Table 3: Comparison of word- and character-based models.

Model	BLEU	chrF
Word-based	0.175	7.99
Character-based	1.247	4.92

Table 4: Comparison of Seq2Seq models with and without attention.

Model	BLEU	chrF
Without attention	0.25	12.0
With attention	5.00	29.6

3.5 Character-based Model

To investigate the effect of input representation, a character-based Seq2Seq model was trained and compared with the baseline word-based model using randomly initialized embeddings. Table 3 summarizes the evaluation results.

The character-based model achieved a higher BLEU score (1.247) than the word-based model (0.175), suggesting that character-level representations can better handle unseen or rare words by modeling subword information. However, the word-based model obtained a considerably higher chrF score (7.99 vs. 4.92), indicating better character-level similarity between the generated translations and the reference sentences.

These results suggest that the two representations have complementary strengths. While character-level modeling improves robustness to vocabulary sparsity, word-level modeling produces more consistent translations according to the chrF metric. Therefore, the choice between word- and character-based models depends on the evaluation criterion and the translation task.

3.6 Attention Model

To improve the baseline Seq2Seq model, a Bahdanau attention mechanism was incorporated into the decoder. The attention model was trained using the same training and evaluation settings as the baseline model, allowing a direct comparison of their performance.

As shown in Figure ??, introducing attention substantially improved translation quality. The attention-based model achieved a BLEU score of approximately 5.0, compared with only 0.25 for the baseline model. Similarly, the chrF score increased from approximately 12.0 to 29.6. These improvements indicate that the attention mechanism enables the decoder to utilize relevant source-side information more effectively instead of relying solely on the encoder’s final hidden state.

Figure 7 shows the training process of the attention model. Both the training and validation losses decreased steadily throughout training, indicating stable optimization without severe overfitting.

To further interpret the model, an attention heatmap is shown in Figure 8. The visualization demonstrates that the decoder assigns higher attention weights to relevant source tokens when generating each target word, confirming that the model learns meaningful source-target alignments.

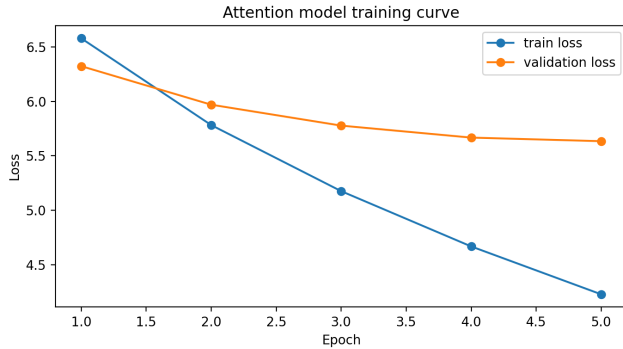


Figure 7: Training and validation loss of the attention model.

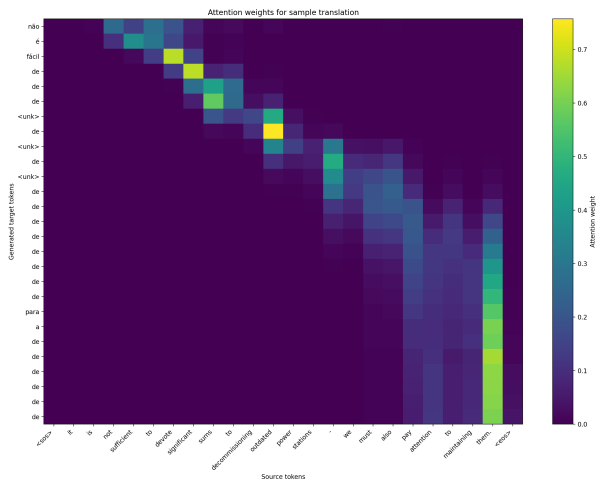


Figure 8: Attention weights for a sample translation.

3.7 Pivot Translation

A pivot translation system was implemented to translate Portuguese into Swedish using English as an intermediate language. Two independent Seq2Seq models were trained: a Portuguese-to-English (PT→EN) model followed by an English-to-Swedish (EN→SV) model. Both models were trained using the same data split, consisting of 21,000 training, 3,000 validation, and 6,000 test sentence pairs.

The PT→EN model achieved a BLEU score of 1.02 and a chrF score of 13.85, while the EN→SV model obtained a BLEU score of 0.35 and a chrF score of 11.20 (Table 5). Figure 9 shows the training and validation losses for both models. The losses decrease steadily during training, indicating stable convergence.

Overall, the PT→EN model outperformed the EN→SV model. Since pivot translation relies on two consecutive translation models, errors introduced during the first translation stage may propagate to the second stage, resulting in lower overall translation quality.

Table 5: Performance of the two component models used for pivot translation.

Model	BLEU	chrF
PT→EN	1.02	13.85
EN→SV	0.35	11.20

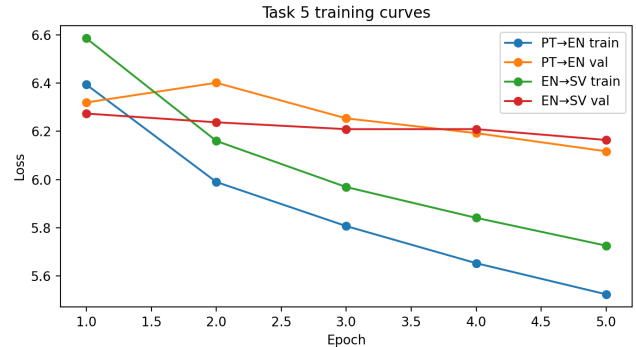


Figure 9: Training and validation loss curves for the PT→EN and EN→SV models.

4 Conclusion

This project investigated English–Portuguese neural machine translation using the Europarl corpus with a recurrent Seq2Seq architecture. Experimental results showed that appropriate preprocessing effectively cleaned the corpus while preserving most sentence pairs for training. Hyperparameter tuning demonstrated that a smaller model with moderate dropout achieved the best baseline performance. Among the embedding methods, pre-trained GloVe-300 embeddings produced the highest translation quality. The translation direction also influenced performance, with the PT→EN model outperforming the EN→PT model in our experiments. Furthermore, the character-based and word-based models exhibited different strengths under different evaluation metrics. The largest improvement was achieved by incorporating the attention mechanism, which substantially increased both BLEU and chrF scores over the baseline Seq2Seq model. Finally, a pivot translation system for Portuguese-to-Swedish translation was successfully implemented, demonstrating the feasibility of multilingual translation through an intermediate language despite error propagation across translation stages.

References

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- [2] Zhixing Tan, Shuo Wang, Zonghan Yang, Gang Chen, Xuancheng Huang, Maosong Sun, and Yang Liu. 2021. Neural machine translation: A review of methods, resources, and tools. *Artificial Intelligence Open* 2 (2021), 10–19. doi:10.1016/j.aiopen.2020.11.001